



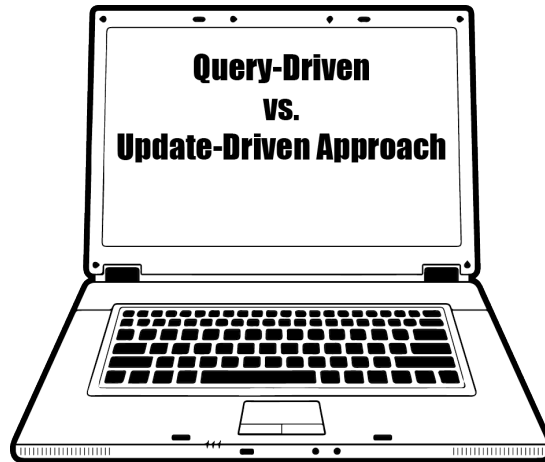
Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!****Integrating Heterogeneous Databases****Learning Outcome:**

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can talk with a colleague about the domain where a warehouse can be implemented.
- I can describe a query-driven approach.
- I can explain update-driven approach.

Ask your partner:

- How do you think a database works?
- Do you think it is important to have an approach to manage database? Why? Why not?



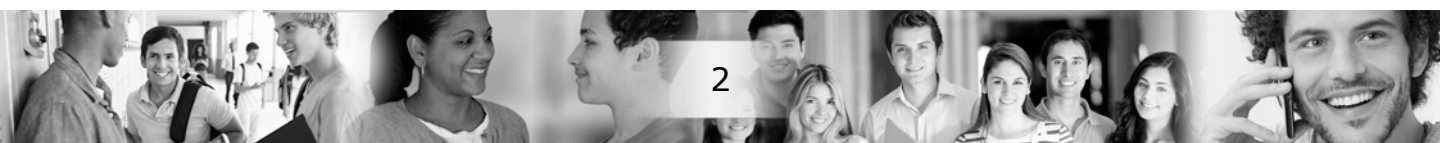
Let's Explore!

Jacquelyn and Blake have to give a conference about data warehousing



Instructions: Read the text below.

- Jacquelyn:** Hey, Blake. Are you nervous about the conference? It's my first time talking in front of so many engineers and I am very anxious.
- Blake:** Don't worry, Jacquelyn. Out of all my colleagues, you are the smartest one. At this point, we know a lot about data warehousing.
- Jacquelyn:** But what do we do if they ask something we don't know?
- Blake:** Mmm, you stumped me. Let's practice a little more.
- Jacquelyn:** Ok, do you know if we can implement data warehouses in any domain?
- Blake:** Well, there are three domains where we can use the information gathered in a warehouse. The first one has to do with tuning production strategies, the second one helps us with customer analysis, and the last one manages everything related to operations analysis.
- Jacquelyn:** Excellent. It's your turn to ask me something.
- Blake:** All right, if we need to integrate heterogeneous databases, what approaches do we have to use?
- Jacquelyn:** We use the query-driven and the update-driven approaches. Those ones are the most common.
- Blake:** Good! If they ask you to describe the query-driven approach, what do you think it's important to mention?
- Jacquelyn:** Well, that approach is common to build wrappers and integrators, which people also call mediators. The process of the query-driven approach includes issuing a query to a client side so that a metadata dictionary can translate it for heterogeneous sites. After that, the query goes to the local query processor and gets into a global answer set.
- Blake:** Great, do you know if that approach has any disadvantages?
- Jacquelyn:** Certainly! It's somehow inefficient, complex, and expensive if we compare it to other approaches. Now it's your turn, Blake. You have to explain the update-driven approach.
- Blake:** In that approach, we integrate the information from multiple heterogeneous sources in a warehouse for direct querying and analysis. The update-driven is capable of providing high performance and it does not require interfaces to process data from local sources.
- Jacquelyn:** I feel better now. We are ready!





Let's Practice!

Characteristics of Two Very Common Approaches

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. This one does not require interfaces to process data from local sources.
2. This one is somehow inefficient, complex, and expensive.
3. This one is capable of providing high performance.
4. This one issues a query to a client side.
5. This one is useful to integrate information from multiple heterogeneous sources.
6. This one is common to build wrappers and integrators.

Query-Driven Approach	Update-Driven Approach

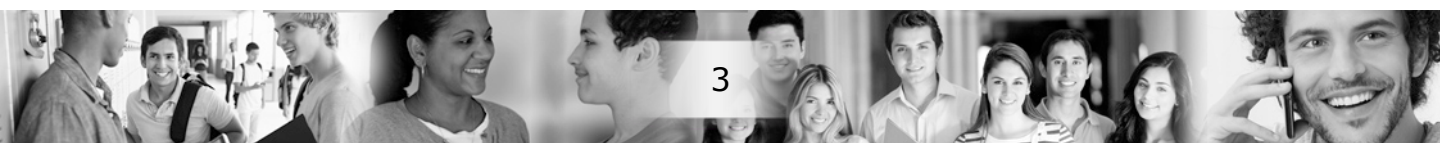


Let's Practice!

There are three domains where we can use the information gathered in a warehouse.

Instructions: Circle the correct word to complete each sentence.

1. The first (one / ones) has to do with tuning production strategies.
2. The second (one / ones) helps us with customer analysis.
3. The last (one / ones) manages everything related to operations analysis.





Extension Activity

Contrasting Approaches

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the query-driven and the update-driven approach.
- Make a comparative chart contrasting those approaches.
- Use one/ones to avoid repeating the name of the approaches.
- Share your chart with the class.

Self-Assessment

Yes No

- I look for more information about the query-driven and the update-driven approach.
- I make a comparative chart to contrast these two approaches.
- I use the pronouns one / ones to avoid repetition.
- I share my chart with the class.

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Target Words

- domain** (noun): a specific area or field
e.g. There are three domains where we can use the information gathered in a warehouse.
- heterogeneous** (adjective): dissimilar or different
e.g. The process of the query-driven approach includes issuing a query to a client side so that a metadata dictionary can translate it for heterogeneous sites.
- processor** (noun): component that interprets and executes signals
e.g. After that, the query goes to the local query processor and gets into a global answer set.

- warehouse** (noun): a place where you can store goods or data
e.g. We integrate the information from multiple heterogeneous sources in a warehouse.
- query** (noun): a search or question
e.g. After that, the query goes to the local query processor and gets into a global answer set.
- source** (noun): the origin of something
e.g. In that approach, we integrate the information from multiple heterogeneous sources in a warehouse for direct querying and analysis.



Learning Points

Zero Conditional in Questions

You can use **zero conditional** to talk about situations that represent facts or rules.

To create questions using zero conditional, you can use the following grammatical structure:

(Wh Word) + Question in Simple Present + **if** + Statement in Simple Present + ?

Examples:

*What do you do **if** you need to use a data warehouse?*

*Does she have to analyze the data **if** she wants to know about customers' preferences?*

Remember that we can reverse the order of the questions and start by using the conjunction **if**; make sure you separate the two clauses with a comma (,).

Examples:

***If** you need to use a data warehouse, what do you do?*

***If** she wants to know about customer preferences, does she have to analyze the data?*

Examples

- *But what do we do if they ask something we don't know?*
- *Do you know if we can implement data warehouses in any domain?*
- *If we need to integrate heterogeneous databases, what approaches do we have to use?*



Learning Points

Pronouns One - Ones

You use the pronouns **one** and **ones** to avoid repeating a noun in a sentence or paragraph.

You use **one** to replace a singular noun when you need to mention it a second time in a sentence or paragraph. You can use **one** after *this*, *that* or an adjective.

Example:

A: *Do I click on the right or left key?*

B: *Click on the left **one**.*

In the previous example, **one** replaces the noun *key*.

You use **ones** to replace plural nouns when you need to mention them a second time in a sentence or paragraph.

Example:

A: *Do you need me to download new programs or update the current **ones**?*

B: *Download new **ones**, please.*

In the previous example, **ones** replace the noun *programs*.

Examples

- *Out of all my colleagues, you are the smartest one.*
- *There are three domains where we can use the information gathered in a warehouse. The first one has to do with tuning production strategies.*
- *We use the query-driven and the update-driven approaches. Those ones are the most common.*

